09 November 2026

MEMORANDUM FOR INDUSTRY PARTNERS

SUBJECT: (U) Request for Information (RFI) on Super Intelligence (SI) Uses for Cislunar Space Vehicle Operations

1. (U) INTRODUCTION

(U) The Department of the Air Force (DAF) is seeking market research information on the potential utility of Super Intelligence (SI) and Machine Learning (ML) algorithms in support of the autonomous operation of space vehicles (SVs) in the cislunar environment. This information may be used to inform the potential development of DAF requirements for future cislunar space systems.

(U) Based on the market information collected with this RFI and all other sources of information relevant to the effort DAF may invest in future studies to mature operationally relevant concepts. Failure to respond to this RFI may result in loss of the ability to participate in future study contracts for this mission area.

(U) This RFI does not constitute a Request for Proposal (RFP), and industry should not construe it as a commitment by the Government. In accordance with Revolutionary FAR Overhaul (RFO) provision 15.101, Early Exchanges with Industry, be advised that this is only an RFI, and does not obligate the Government to reimburse any cost associated with preparation and/or submission of a RFI response. This RFI does not guarantee that the Government will issue an RFP or award a contract to provide these supplies or services.

2. (U) BACKGROUND

(U) The Government has an over-arching policy objective to develop a sustained architecture beyond earth orbit. In support of this policy objective, DAF is investigating concepts and systems to support the implementation of cislunar operations in the DAF baseline. Operation of cislunar systems is challenged by the orders of magnitude increase in the cislunar domain volume, relative to the near-earth (LEO/MEO/GEO) domain volume. Communication latencies that result from the increase in volume may mean that traditional, ground-based, command and control and mission planning systems are no longer scalable on operationally relevant timelines.

3. (U) PURPOSE

(U) The Government is seeking to identify respondents with relevant cislunar operations and super intelligence domain expertise whose RFI response provides the Government with the opportunity to review and assess existing, flight-demonstrated super intelligence and machine learning software products capable of near-term adaptation and integration for cislunar operations, to inform future requirements for cislunar systems and operations. The Government is interested in innovative, flight-ready, concepts to enable cislunar SVs to perform autonomous station-keeping, trajectory optimization, and hazard avoidance in non-Keplerian regimes. The Government is interested in approaches across the spectrum of SI integration, from learned components embedded within physics-based guidance, navigation, and control (GNC) functions to SI-enabled mission-level orchestration of onboard autonomy.

(U) The Government is interested in existing, flight-ready autonomy software with a preference for commercial off-the-shelf (COTS) and/or near-COTS software capabilities that would require no new model development and minimal training or adaptation. Government schedule objectives would require initial software delivery within ninety (90) days of Authority to Proceed (ATP) and fully qualified flight software by ATP plus eighteen (18) months. Initial software delivery shall consist of the respondent's existing executable software baseline and its validation in a software-in-the-loop environment; platform-specific integration and flight qualification may follow within the overall eighteen (18) month schedule. Respondents should assume that the software stack would be integrated into Government furnished SV hardware developed on a separate effort. Prior cislunar flight heritage is desirable but not required: flight heritage in other orbital regimes is acceptable where the respondent demonstrates applicability of the software to cislunar dynamics and operations.

(U) The Government is interested in understanding industry's ability to rapidly deliver existing software products to support this mission area. It is not interested in investing in a cislunar SI research and development program.

4. (U) TECHNICAL FOCUS AREAS

(U) Whitepapers should address one or more of the following core technical focus areas:

1. (U) Autonomous Mission Management: SI-enabled interpretation of mission objectives and onboard data to plan, sequence, or re-task vehicle actions and sustained management of vehicle subsystems to remain safely operable during extended periods without ground contact;

2. (U) Predictive Maintenance and Fault Isolation: On-board anomaly detection systems that can dynamically respond to unexpected and/or ambiguous hardware faults and degradation;

3. (U) On-board Trajectory Optimization: Autonomous maneuver planning and propulsion management capable of recalculating optimal transfers without ground-station intervention;

4. (U) On-board computer vision: multi-model image processing and identification to include identification of celestial bodies and man-made objects

5. (U) Other areas the respondent has experience with that may be of operational utility for cislunar operations.

(U) The Government is not interested in Respondent's approach to hardware design/development to support the cislunar mission area. It does, however, need to understand the hosting hardware requirements necessary to implement Respondent's software solution. Whitepapers should provide discussion of the compute resources that may be required to implement the SI/ML algorithms/models used to address the technical focus areas. Specifically:

> 1. (U) Processor Architecture: Discuss required compute hardware, for example radiation hardened FPGAs, multi-core CPUs, system-on-chip ASICs, and/or neuromorphic processors;
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> 2. (U) Compute Performance: Discuss estimated peak performance metrics (Floating Point Operations per Second, Tera Operations per Second, etc.);

3. (U) Memory Allocation: Discuss the required memory footprint, including minimum volatile memory (RAM) for runtime execution and non-volatile memory (Flash, NAND) for model storage;

4. (U) Power Requirements: Discuss the potential power requirements for the processing architecture during average and peak power regimes;

5. (U) Software Stack: Discuss the inference runtime and software framework required to execute the SI/ML models, expected model size and quantization approach, and any dependence on specialized libraries or accelerator-specific toolchains;

6. (U) Model Update Concept: Discuss how trained models would be updated, retrained, or rolled back on orbit, including expected uplink volume for a model update, onboard validation performed before an updated model is permitted to influence vehicle operations, and fallback behavior if an updated model fails validation.

7. (U) Model Scale and Power: State the expected model scale (parameter count) and estimated inference power draw for each SI/ML component, including any duty-cycling approach required to operate large models within the processing architecture's power budget;

(U) Whitepapers should discuss the data required to develop and validate the proposed approach. Specifically: the training and validation data or simulation environments required, the respondent's approach to bridging simulation and flight conditions (sim-to-real), whether Government-furnished data or models would be used or required, and any anticipated data rights considerations.

(U) Whitepapers should discuss the approach to verification and validation of autonomous behaviors including: enforcement of safety bounds or operational constraints on model outputs, fallback or safe-mode behavior when the model cannot produce a trusted output, expected behavior under sensor dropout or degraded inputs, runtime monitoring of model performance, and robustness to off-nominal or adversarial inputs.

(U) Whitepapers should identify any functions that assume ground reach-back or off-board processing, and describe the concept's behavior when those communication links are unavailable.

5. (U) RFI SUBMITTAL INSTRUCTIONS

(U) Interested respondents or respondent teams should submit a response of no more than five (5) pages that includes the following information:

1. (U) A cover page with:

    a. (U) Company name;

    b. (U) A primary and secondary Point of Contact (POC) for the company, to include name, telephone number, and email address to coordinate any additional communication;

    c. (U) Submittal date for the RFI response.

2. (U) Description of recent (within the past three (3) years) relevant respondent experience in the following areas:

a. Development and on-orbit operation of autonomous spacecraft flight software, identifying the orbital regimes in which the respondent has experience;

b. Design, development, and operation of classified space systems (preferred, not required);

c. Design, development, and application of Super intelligence / Machine Learning (SI/ML) to space operations.

3. (U) The proposed approach for addressing one or more of the focus areas described above. Include discussion of the following:

a. Details of the technical approach to addressing the chosen focus area(s), including how learned components interface with physics-based estimation, guidance, and control functions;

b. Assessment of the maturity of the technical approach, including a discussion of the development path for implementation in a space-based demonstration or operational space system;

c. Compute resources required to execute the approach, stated as the hosting requirements the software places on Government furnished compute; respondents are not being asked to design or supply compute hardware;

d. Prior demonstrations of the proposed software, identifying: the product and version demonstrated; the flight or customer deployment; the orbital regime; the processor and sensor interfaces used; whether operation was advisory or closed-loop; and the new development, if any, required for the proposed demonstration.

4. (U) ROM cost estimate for licensing, adaptation, integration, and demonstration of Respondent's existing software capability consistent with Government schedule objectives..

5. (U) A credible, high-level schedule for delivery, validation, and flight qualification of the existing software capability, consistent with Government schedule objectives.

A. (U) General Instructions

(U) Interested respondents shall appropriately classify RFI responses. Interested respondents shall limit responses to no more than five (5) pages, including supporting appendices, cover page, indices, etc. Security classification, portion marking, and company sensitive marks are required. Type size shall be 12-point proportional font. Briefings and/or briefing charts in lieu of written responses are not permitted.

(U) Responses shall be sent electronically to the following POCs no later than **1200 noon EST 04 December 2026**.

- Mr. Shelby Sommer (Shelby.Sommer@us.af.mil)
- Lt Col Joseph DeRienzo (Joseph.DeRienzo@us.af.mil)

(U) Responses to this notice shall include Respondent's Company Name, Address and Contact Information. Responses should be prepared in Microsoft Word 2016 (or compatible version) or PDF format (compatible with Adobe Reader XI). When submitting a response, please use the subject "DAF Cislunar Space SI – [Respondent Name]". Any responses submitted through different channels or to different email addresses will not be considered. The Government is not obligated to review responses to this RFI received after the 04 December 2026 deadline.

(U) If a respondent would like to include classified information that cannot be submitted on NIPR, or would like to submit their response via another network (e.g., JWICS, SIC or CORE), please contact the POCs listed above for submission instructions. Please indicate the classification level desired.

6. (U) CONSENT TO DISCLOSURE AND COMPANY-TO-COMPANY NON-DISCLOSURE AGREEMENTS (NDAS)

(U) Pursuant to RFO 9.505-4(b), Obtaining Access to Proprietary Information, a contractor that gains access to proprietary information of other companies in performing A&AS for the Government must agree with the other companies to protect their information from unauthorized use or disclosure for as long as it remains proprietary and refrain from using the information for any purpose other than that for which it was furnished. The Contracting Officer (CO) shall obtain copies of these agreements and ensure that they are properly executed.

(U) To the maximum extent possible, please submit non-proprietary information. If proprietary information is submitted, it must be appropriately and specifically marked. It is the submitter's responsibility to clearly define to the Government what is considered proprietary data. Any proprietary information should be clearly labeled as "Proprietary." The Government will not publicly disclose proprietary information obtained as a result of the RFI. To the full extent that it is protected pursuant to the Freedom of Information Act and other laws and regulations, information properly identified by a respondent as "Proprietary" will be appropriately controlled.

(U) Respondents are hereby notified that the Government intends to utilize non-Government consultants, e.g., Systems Engineering and Technical Assistance (SETA) contractors and Federally Funded Research and Development Center (FFRDC) personnel, to review RFI responses and provide technical advice. All personnel assigned to provide technical advice have signed non-disclosure agreements and conflict of interest statements and will be made aware that RFI responses may not be duplicated, used, or disclosed in whole or in part for any purpose other than evaluation of the responses.

7. (U) GOVERNMENT REVIEW OF RESPONSES

(U) The Government intends to review each RFI response with a focus on the following risk areas to determine the ability of the marketplace to meet the Government's technical and schedule objectives:

a. (U) Technical implementation risk with each respondent or respondent team's technical approach and Concept of Operations (CONOPS) for addressing the technical focus areas. This includes the risk associated with the technical maturity of the proposed approach.

b. (U) Risk associated with the schedule for delivery of validated flight qualified software.

c. (U) Risk associated with the respondent's prior domain expertise in cislunar SV operations and implementation of SI/ML models and algorithms for space-based applications.

d. (U) Respondent or respondent team's overall ability to execute the program.

e. (U) Risk associated with the maturity of the SI/ML models, algorithms, and software, considering prior demonstrations in simulation, against recorded flight data, on flight-representative processors, and on orbit in advisory or closed-loop modes.

(U) Responses to this RFI will not be returned. Intellectual or other privileged or proprietary information contained in responses to this RFI will not be distributed outside of the Department of the Air Force or to other Government employees from other Government agencies. Respondents are advised that the Government is under no obligation to acknowledge receipt.

(U) The Government does not intend to provide feedback to RFI respondents; however, responses may lead to an invitation by the Government for further discussion. Failure to respond to this RFI may result in loss of the ability to participate in subsequent studies based upon Government consideration of all sources of information relevant to this effort.

(U) Based on the results of this RFI the Government intends to focus further engagement with respondents that demonstrate a mature approach using existing, flight-proven software. The Government reserves the right to potentially issue multiple contracts/agreements based on our findings. Further, the Government may solicit multiple rounds of contracts to reduce risk and mature technologies.

8. (U) QUESTIONS AND POINTS OF CONTACT

(U) Respondents may submit questions concerning this RFI NLT **1200 noon EST 21 November 2026** via NIPR to the following POCs:

- Mr. Shelby Sommer (Shelby.Sommer@us.af.mil)
- Lt Col Joseph DeRienzo (Joseph.DeRienzo@us.af.mil)

(U) The Government may answer questions at its discretion on a case-by-case basis.

(U) Responses shall be sent electronically to the following POCs no later than **1200 noon EST 04 December 2026**.

(U) The Government would like to thank all prospective respondents in advance for expending time and resources in support of this RFI.